



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: DATA ANALYTICS AND VISUALIZATION (24DSB3101)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	Retail Sales Analytics and Customer Intelligence Framework Using Machine Learning and Business Intelligence	
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Abstract

The retail industry generates a vast amount of transactional data, but traditional reporting systems often provide limited insights for effective business decision-making. This project proposes a **Retail Sales Analytics and Customer Intelligence Framework Using Machine Learning and Business Intelligence** to transform raw retail data into actionable insights. The **Superstore Sales Dataset** is preprocessed using Python, NumPy, and Pandas, followed by EDA and statistical analysis using Matplotlib and Seaborn to identify sales, customer, product, and regional patterns. Machine learning models including **Linear Regression, Decision Tree, Random Forest, XGBoost, and K-Means Clustering** are applied for sales prediction and customer segmentation, with **XGBoost as the primary model for sales prediction**. The framework further introduces **Customer Lifetime Value (CLV) Prediction** using RFM-based features to estimate future customer value and **Scenario-Based Business Analysis** to evaluate the impact of variables such as discount and quantity on predicted sales. The processed data is stored in **MySQL**, while **Power BI** provides interactive dashboards for sales, customer, product, regional, CLV, and scenario-based insights.

Signature of the Guide

HOD